

Junkun Liu

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EDUCATION

Johns Hopkins University

B.S. in Applied Mathematics & Statistics and Computer Science | GPA: 3.74/4.00

Baltimore, MD

Expected May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Mathematical Foundations of Computer Science, Theory of Computation, Probability, Mathematical Statistics, Optimization, Linear Algebra, Full-Stack JavaScript

EXPERIENCE & PROJECTS

LaunchStack [\[GitHub\]](#)

Baltimore, MD / Remote

Tech Lead, Founder Operating System

Jun 2026 – Present

Product Engineering Intern

Spring 2026

- Lead a **product engineering team**, owning technical architecture, task decomposition, implementation sequencing, and **code/PR review** across the Founder Operating System workstream.
- Designed the **end-to-end document-change pipeline**: fetch versioned artifacts, load historical chunks, deterministically align adjacent versions, compute structured **document_change** evidence, and preserve source IDs/provenance before downstream synthesis.
- Architected the **Founder Weekly Review / RAG pipeline**, aggregating documents, GitHub activity, customer feedback, and team context into source-grounded evidence snapshots used to generate changes, blockers, customer signals, and next priorities.

Personal Action Agent [\[GitHub\]](#) | *Independent AI Systems Project*

Jun 2026 – Present

- Architected an end-to-end **agent harness** that converts bidirectional Gmail evidence into durable operational Actions through provider-neutral normalization, selective conversation hydration, **typed structured outputs**, prompt/semantic contracts, deterministic policy gates, migration-safe SQLite state, and append-only provenance.
- Built a reliability boundary where the **LLM interprets semantics but cannot directly mutate state**; deterministic code owns identity, lifecycle transitions, idempotency, evidence-completeness checks, and persistence authorization.
- Built a **systematic agent-evaluation harness** with 275 regression tests, human-reviewed Gmail gold data, structured-output conformance checks, and prompt/semantic error analysis; using results to redesign multi-process ownership semantics and extend **retrieval/RAG** over persistent evidence.

StepFun

Beijing, China

Algorithm Engineer Intern

Jun 2025 – Aug 2025

- Built and evaluated **LLM-based web agents** for multi-step browser tasks, testing completion stability under tool coordination, navigation, and dynamic DOM changes.
- Integrated **structured agent workflows** and Azure OpenAI-compatible model access into training/evaluation pipelines; containerized browser environments and diagnosed navigation loops, tool misuse, element ambiguity, and inconsistent state transitions.

RESEARCH

Dynamic Taint Analysis Research

Research Collaboration

Second-listed Author, Manuscript in Preparation

Jun 2026 – Jul 2026

- Coauthored a manuscript on **dynamic taint analysis across CPython, SpiderMonkey, and Chromium**; implemented propagation through function calls, locals/parameters, property/index access, string concatenation, and native string operations.
- Built reproducible **full-application CVE validations** tracing tainted inputs to DOM and code-execution sinks; analyzed taint-propagation DAGs, source-to-sink coverage, propagation gaps, and runtime limitations.

LEADERSHIP & TEACHING

Johns Hopkins University PILOT Program

Baltimore, MD

Head PILOT Leader, Honors Discrete Mathematics

Jul 2026 – Present

PILOT Leader, Honors Discrete Mathematics

Jul 2025 – Jun 2026

- Promoted to **Head PILOT Leader** after a year of leading weekly sessions; oversee the Honors Discrete Mathematics PILOT group and lead problem-solving sessions for 10–15 students on proofs, logic, induction, set theory, and combinatorics.

Johns Hopkins University

Baltimore, MD

Incoming Teaching Assistant, Mathematical Foundations of Computer Science

Fall 2026

- Selected as an incoming **Teaching Assistant** for the proof-based CS core course after earning an A+; appointment begins Fall 2026.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL

AI/ML: LLM Agents, Agent Harnesses, RAG, Prompt Engineering, Structured Outputs, Agent Evaluation, Embeddings, PyTorch, Scikit-learn

Systems/Research: SQLite, State Machines, Provider Abstractions, Dynamic Taint Analysis, Browser Automation, CVE Validation, Benchmarking, Docker

Web/Data: React, Next.js, Node.js, PostgreSQL, Git, AWS